



The City College
of New York

CSC 59867-E: Senior Project II

AI Agents for Decision Making in the Real World

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Recap on this Semester's Focus

We are transitioning from the algorithm exploration and prototyping phase of Senior Project I to rigorous physical and virtual implementation in Senior Project II.

Recall that Generative AI relies on passive probability generation localized in historical text.

AI Agents act as autonomous decision engines that actively perturb their environment and receive feedback.

We must account for how poorly parameterized reward functions incentivize policies to exploit metric loopholes rather than achieving true task objectives.



Logistics and Reminders

Federal and CUNY student aid compliance requires in-person signature verification on the attendance roster.

We are reallocating traditional homework hours into 15–30 minute in-class engineering sprints to address open-ended, non-automatable implementation challenges.

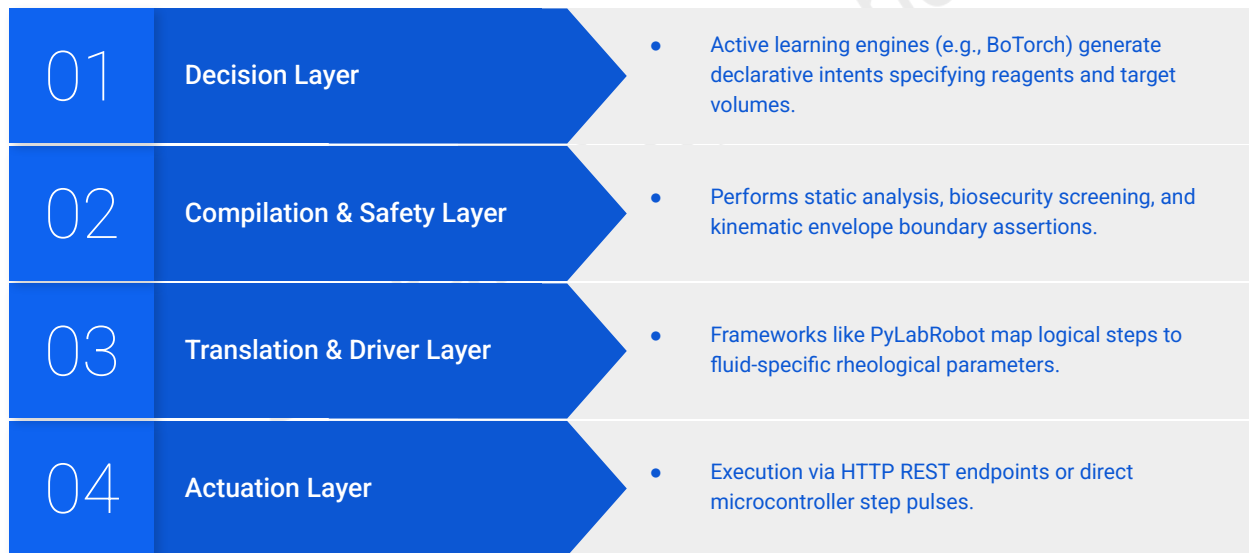
Students needing access to the Empire AI cluster (featuring B200 GPUs) must submit their phone number and personal Gmail to interface with the HPCC director.

Due to campus email migration network disconnects, SMS validation is required within 24 hours.

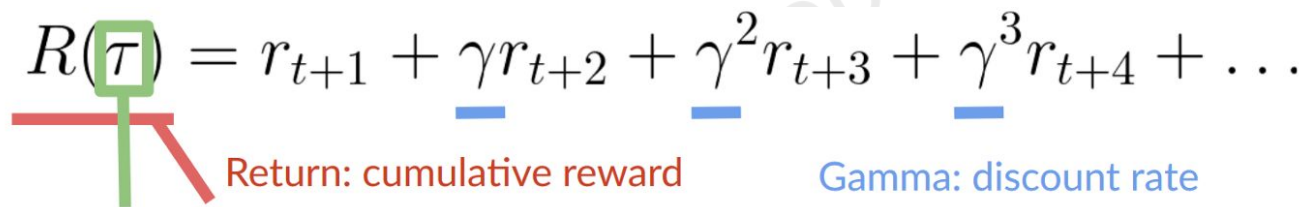


Four Tiers of Autonomous Architecture

Transitioning AI from simulation to the physical lab requires decoupling high-level intent from low-level joint-space trajectories:



Recall: Reinforcement Learning



The diagram shows the equation $R(\tau) = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \gamma^3 r_{t+4} + \dots$. A green box highlights the τ in $R(\tau)$. A red line points from the box to the text "Return: cumulative reward". A blue line points from the γ in the first term to the text "Gamma: discount rate".

$$R(\tau) = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \gamma^3 r_{t+4} + \dots$$

Return: cumulative reward

Gamma: discount rate

Trajectory (read Tau)
Sequence of states and actions

$$R(\tau) = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}$$



Formalization of the Sim2Real Gap

Deploying policies trained in physics simulators to real manipulators exposes the Sim2Real gap (driven by joint compliance, contact dynamics, and optical properties).

Let $\xi \in \Xi$ encapsulate unmodeled physical parameters (friction, mass, lighting). Domain Randomization optimizes the policy π_θ across a parameter distribution $p(\xi)$:

$$\theta^* = \arg \max_{\theta} \mathbb{E}_{\xi \sim p(\xi)} \left[\mathbb{E}_{\tau \sim \pi_\theta, \mathcal{P}_\xi} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t) \right] \right]$$

System Identification (SysID) centers this distribution by minimizing empirical dynamics residuals from physical trajectories.



Visuomotor Control

The HuggingFace Robotics course and [lerobot](#) repository provide standardized toolkits for deploying Vision-Language-Action (VLA) models onto physical manipulators.

Action Chunking with Transformers (ACT): Mitigates compounding physical errors by predicting a sequence of k future actions $\mathbf{A}_{t:t+k}$ simultaneously.

The Conditional Variational Autoencoder (CVAE) loss minimizes action reconstruction error while regularizing the latent posterior against a Gaussian prior via KL divergence:

$$\mathcal{L}_{\text{ACT}} = \mathbb{E} \left[\sum_{j=0}^{k-1} \|a_{t+j} - \hat{a}_{t+j}\|_1 + \beta D_{\text{KL}}(q_{\phi}(z|\mathbf{A}, o_t) \| \mathcal{N}(\mathbf{0}, \mathbf{I})) \right]$$



Real-Time Control

Physical robotic manipulation requires deterministic temporal loops. Standard laboratory arms operate at 50Hz, enforcing a strict cycle deadline of $\Delta t = 20\text{ms}$.

Latency Budget Breakdown:

- Observation Acquisition (Camera to Buffer): 8.0--10.0ms
- Visual Feature Extraction (ResNet/ViT): 3.0--5.0ms
- Policy Chunk Evaluation (Transformer Decoder): 3.0--6.0ms
- Serial Bus UART Packet Serialization: 2.0--4.0ms

If neural network inference latency exceeds 20ms the robot experiences setpoint starvation, leading to high-frequency mechanical vibration.



Hardware Abstraction Layers (HALs)



1. Hardware Agnostic

High-level agent algorithms must remain hardware-agnostic; agents should emit declarative experimental intents.



2. Task Specifications

Experiments are defined as validated JSON/YAML task specifications specifying physical identities, target volumes, and liquid classes.



3. Validation & Compilation

The HAL validates schemas against kinematic bounding boxes and deck clearance margins before compiling them into Opentrons API calls.



Kinematic Envelope Verification

To prevent mechanical collisions and equipment damage, the HAL layer parses the declarative schema against physical hardware limits.

Cartesian Workspace Boundaries: All target positions must fall within physical limit switches:

$$x \in [x_{\min}, x_{\max}], \quad y \in [y_{\min}, y_{\max}], \quad z \in [z_{\min}, z_{\max}]$$

Deck Clearance Assertions: During traversal, the pipette must retract to a verified safe height:

$$z_{\text{clear}} \geq \max_k(h_{\text{labware},k}) + \Delta z_{\text{margin}} \quad \text{to avoid lateral collisions with thermal cyclers or tall reservoirs.}$$



Fluid Dynamics for Autonomous Liquid Handling

Factory robotic calibrations assume pure water ($\mu \approx 1.0 \text{ mPa} \cdot \text{s}$). Autonomous screening introduces volatile organic solvents and high-viscosity polymers.

Laminar intake through a conical pipette tip is governed by the Hagen-Poiseuille relationship:

$$\Delta P = \frac{8\mu LQ}{\pi R(z)^4}$$

If agents demand excessive volumetric flow rates (Q) for highly viscous fluids (μ), ΔP exceeds the air column's limit, causing cavitation voids. Liquid classes parameterize specific aspiration rates and dwell delays to stabilize the interface.



Class Discussion Section (4 Minutes)

Scenario:

Your AI agent specifies an automated liquid transfer of 50 microL of hot Dimethyl Sulfoxide (DMSO) on the Opentrons Flex.

Task:

Pair up and discuss (4 minutes). What fluid mechanical mitigations must the HAL inject for a highly volatile liquid class compared to a standard aqueous buffer?

Guiding Questions:

How do elevated vapor pressures impact the air gap inside the pipette? How can we prevent unprompted dripping during gantry transit?



OpenTrons Autonomous Experimentation

Python Protocol API (v2/v3): An imperative paradigm operating on the robot's embedded runner. Best for static, predefined experimental sequences. We would use AI agents to dynamically handle experiment parameters beforehand.

HTTP REST API (Port 31950): Essential for online, reactive AI agents that can dynamically take decisions

The REST API utilizes a stateful transaction model:

- `POST /runs` initializes a session.
- `POST /runs/{run_id}/commands` streams atomic actions (e.g., `aspirate`, `dispense`, `moveTo`).
- `GET /runs/{run_id}/commands` captures hardware telemetry and error codes.



OpenShelf Storage and Retrieval

While Opentrons handles micro-volumes, Automated Storage and Retrieval Systems (AS/RS) like OpenShelf manage macro-scale lab inventory logistics.

Schedulers dispatch an authenticated `POST /api/v1/retrieve` request specifying a sample UUID. The AS/RS verifies the 2D DataMatrix code, logs mass via an internal gravimetric scale, and ejects the plate.

Queue Latency Management: Mechanical extraction requires $t_{\text{retrieval}} = 20 \pm 5$ seconds. Agents must pre-fetch containers to overlap logistics latency with ongoing liquid handling steps.



Sim2Real Physical Alignment (Coordinate Registration)

When we're handling pipettes, glass beakers, and other fragile materials needed for peptide sequencing using the OpenShelf robots, we need sub-mm precision, achieved with the transformation matrix $\mathbf{T}_B^S \in SE(3)$

The optimal spatial alignment between the arm's base $\{B\}$ and the shelf port $\{S\}$ is solved via the orthogonal Procrustes problem:

$$\mathbf{R}_B^S, \mathbf{t}_B^S = \arg \min_{\mathbf{R} \in SO(3), \mathbf{t} \in \mathbb{R}^3} \sum_{i=1}^N \|\mathbf{p}_{B,i} - (\mathbf{R}\mathbf{p}_{S,i} + \mathbf{t})\|_2^2$$

Evaluating the Singular Value Decomposition (SVD) of the cross-covariance matrix $\mathbf{H} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$ yields the closed-form rotation $\mathbf{R}_B^S = \mathbf{U}\mathbf{V}^T$.



Scalable Active Learning with Bayesian Optimization

Active learning combines every step of the scientific process into an iterative cycle to continuously improve results.

Evaluating physical samples serially wastes robotic system resources.

We use multi-point Expected Improvement (q -EI) to evaluate batches of size q simultaneously.

The batch acquisition function is optimized using Monte Carlo integration and the reparameterization trick over the Cholesky decomposition of the predictive covariance:

$$\hat{\alpha}_{q\text{-EI}}(\mathbf{X}) = \frac{1}{M} \sum_{m=1}^M \max \left(0, \max_{i=1 \dots q} y_i^{(m)}(\mathbf{X}) - y^* \right)$$

Questions?

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